

Multi-Agent System: Theories and Applications

Lecture 4: Backgrounds on Graph Theory

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Outline

- 1 Undirected and Directed Graphs
- 2 Graph Connectivity
- 3 Graph Periodicity
- 4 Algebraic Properties

Definition and examples

What is a graph?

A graph is a mathematical structure which is used to model pairwise relations between objects. Each object is called a *node (or vertex)* and the relation between two nodes is represented by an *edge*. A graph is usually denoted by $\mathcal{G}$ with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, or in short $(\mathcal{G}, \mathcal{V}, \mathcal{E})$.

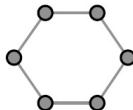
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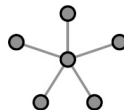
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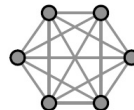
(a) *Path graph* with 6 nodes, denoted by P_6



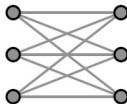
(b) *Cycle (ring) graph* with 6 nodes, denoted by C_6



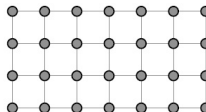
(c) *Star graph* with 6 nodes, denoted by S_6



(d) *Complete graph* with 6 nodes, denoted by K_6



(e) *Complete bipartite graph* with 3+3 nodes, denoted by $K_{3,3}$



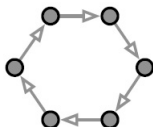
(f) A Cartesian grid graph



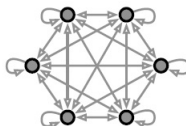
(g) *Petersen graph*

Direction of edges: undirected vs. directed

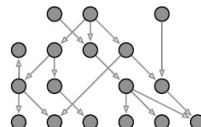
A graph is called *undirected* if its vertices are connected by *unordered* edges. If the edges are *ordered*, then the graph is called *directed graph*, or *digraph*.



(a) *Cycle digraph* with 6 nodes



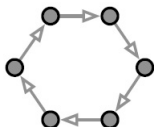
(b) *Complete digraph* with 6 nodes



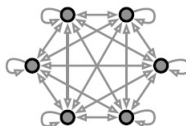
(c) A digraph with no directed cycles

Direction of edges: undirected vs. directed

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(a) *Cycle digraph* with 6 nodes



(b) *Complete digraph* with 6 nodes



(c) A digraph with no directed cycles

Note: undirected graphs are special cases of digraphs

Neighbor and degree

Undirected graph

- Any vertex connected to a vertex is called its *neighbor*. The neighboring set of a vertex i is denoted by $\mathcal{N}_i$.
- The number of neighbors of a vertex is called its *degree*, denoted by d_i .

Neighbor and degree

Undirected graph

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Digraph

- For an ordered edge, which starts from vertex u to vertex v , u is called the parent vertex and v is called the child vertex. Moreover, v is called an *out-neighbor* of u and u is called an *in-neighbor* of v .
- The sets of in-neighbors and out-neighbors of a vertex i are denoted by $\mathcal{N}_i^{\text{int}}$ and $\mathcal{N}_i^{\text{out}}$; respectively.
- The numbers of out-neighbors and in-neighbors of a vertex is called its *out-degree* and *in-degree*; respectively.

Connectivity of undirected graphs

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is a sequence of consecutive edges. Another name is *walk*. A path is *simple* if no vertex appears more than once in it.

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Connectivity

An undirected graph is *connected* if there is a path between any two vertices.

- A graph having no cycle is called *acyclic*.
- A connected acyclic graph is called a *tree*.

For digraphs

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Directed spanning tree

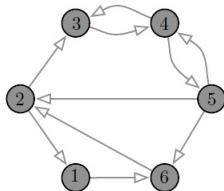
is a directed tree having the same vertex set with the digraph.

Connectivity of digraphs

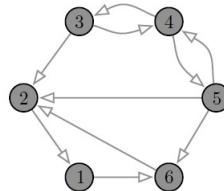
- $\mathcal{G}$ is *strongly connected* if there exists a directed path from any node to any other node.
- $\mathcal{G}$ is *weakly connected* if its undirected version is connected.
- $\mathcal{G}$ has a *globally reachable node* if there is a directed path from any other node to it.

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(a) A strongly connected digraph



(b) A weakly connected digraph with a globally reachable node

For strongly connected digraphs

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- Any strongly-connected digraph with a self-loop is aperiodic.
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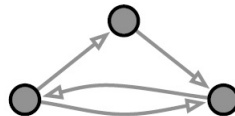
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(a) A periodic digraph with period 2



(b) An aperiodic digraph with cycles of length 1 and 2.



(c) An aperiodic digraph with cycles of length 2 and 3.

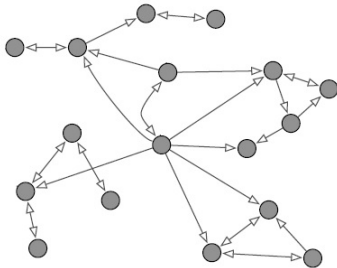
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How about this graph?

Graph matrices

Adjacency matrix

For a digraph $\mathcal{G}$ having n vertices, its adjacency matrix is denoted by $\mathcal{A} \triangleq [a_{ij}] \in \mathbb{R}^{n \times n}$, where a_{ij} is a *weight* on the edge connecting vertex j to vertex i ; $a_{ij} = 0$ if vertices i and j are not connected.

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Degree matrix

- In-degree matrix: $\mathcal{D}^{\text{in}} \triangleq \text{diag}(d_i^{\text{in}})_{i=1,\dots,n}$, where $d_i^{\text{in}} \triangleq \sum_{j \in \mathcal{N}_i^{\text{in}}} a_{ij}$
- Out-degree matrix: $\mathcal{D}^{\text{out}} \triangleq \text{diag}(d_i^{\text{out}})_{i=1,\dots,n}$, where $d_i^{\text{out}} \triangleq \sum_{j \in \mathcal{N}_i^{\text{out}}} a_{ji}$
- A graph is *balanced* if $d_i^{\text{in}} = d_i^{\text{out}} \forall i = 1, \dots, n$

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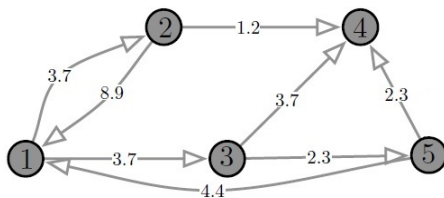
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Laplacian matrix

Denoted by $\mathcal{L} \triangleq \mathcal{D}^{\text{in}} - \mathcal{A}$.

Example

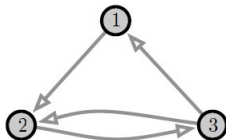


$$\mathcal{A} = \begin{bmatrix} 0 & 8.9 & 0 & 0 & 4.4 \\ 3.7 & 0 & 0 & 0 & 0 \\ 3.7 & 0 & 0 & 0 & 0 \\ 0 & 1.2 & 3.7 & 0 & 2.3 \\ 0 & 0 & 2.3 & 0 & 0 \end{bmatrix} \Rightarrow \mathcal{L} = \begin{bmatrix} 13.3 & -8.9 & 0 & 0 & -4.4 \\ -3.7 & 3.7 & 0 & 0 & 0 \\ -3.7 & 0 & 3.7 & 0 & 0 \\ 0 & -1.2 & -3.7 & 7.2 & -2.3 \\ 0 & 0 & -2.3 & 0 & 2.3 \end{bmatrix}$$

Relation with graph connectivity and periodicity

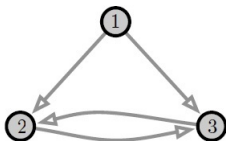
Adjacency matrix vs. graph connectivity

A digraph $\mathcal{G}$ with $n \geq 2$ vertices is strongly connected $\iff$ its adjacency matrix $\mathcal{A}$ is irreducible



$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

irreducible



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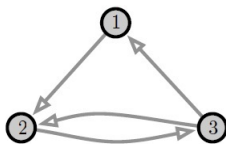
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Relation with graph connectivity and periodicity

Adjacency matrix vs. graph periodicity

A digraph $\mathcal{G}$ with $n \geq 2$ vertices is strongly connected and aperiodic $\iff$ its adjacency matrix $\mathcal{A}$ is primitive



$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

primitive



Spectrum of adjacency matrix

For a digraph $\mathcal{G}$ with non-negative adjacency matrix $\mathcal{A}$,

$$\min \mathcal{A} \mathbf{1}_n \leq \rho(\mathcal{A}) \leq \max \mathcal{A} \mathbf{1}_n$$

where $\rho(\mathcal{A})$ is the spectral radius of $\mathcal{A}$, $\mathbf{1}_n$ is the vector having n elements equal to 1.

Spectrum of Laplacian matrix

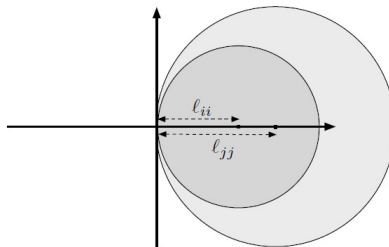
- $\mathcal{L}$ is a right stochastic matrix, hence 0 is one eigenvalue of $\mathcal{L}$ with eigenvector $\mathbf{1}_n$.
- $\mathcal{L}$ has a single eigenvalue 0 if and only if $\mathcal{G}$ has a spanning tree.
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Spectrum

All non-zero eigenvalues of $\mathcal{L}$ have strictly positive real parts (Hint: Gershgorin Theorem).



Algebraic connectivity (only for undirected graphs)

Eigenvalues of $\mathcal{L}$ are: $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Algebraic connectivity

is the second smallest eigenvalue of $\mathcal{L}$, i.e., λ_2 . Also called the Fiedler eigenvalue.

- $\mathcal{G}$ is connected if and only if $\lambda_2 > 0$.
- The multiplicity of eigenvalue 0 is equal to the number of connected components of $\mathcal{G}$.
- If $\mathcal{A} \leq \tilde{\mathcal{A}}$, then $\lambda_2(\mathcal{A}) \leq \lambda_2(\tilde{\mathcal{A}}) \rightarrow$ effect of edge weights and graph dense to λ_2

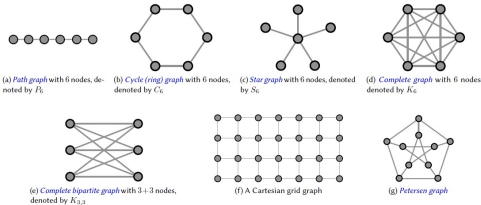
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Graph

path graph P_n

cycle graph C_n

star graph S_n

complete graph K_n

complete bipartite $K_{n,m}$

Algebraic Connectivity

$$\lambda_2(P_n) = 2(1 - \cos(\pi/n))$$

$$\lambda_2(C_n) = 2(1 - \cos(2\pi/n))$$

$$\lambda_2(S_n) = 1$$

$$\lambda_2(K_n) = n$$

$$\lambda_2(K_{n,m}) = \min(n, m)$$